



Terraform Syllabus: From Basic to Advanced

1. Introduction to Terraform

- What is Terraform?
- Infrastructure as Code (IaC) Concepts
- Terraform vs. Other IaC Tools (Ansible, CloudFormation, Pulumi)
- Terraform Use Cases
- Terraform Workflow (Write → Plan → Apply)

2. Setting Up Terraform

- Installing Terraform (Linux, macOS, Windows)
- Configuring Terraform CLI
- Terraform Providers Overview
- Authentication with Cloud Providers (AWS, Azure, GCP)

3. Terraform Basics

- Terraform Configuration Files (.tf)
- Terraform Block (terraform {})
- Provider Block (provider "aws" {})
- Resource Block (resource "aws_instance" "web" {})
- Variables & Outputs
- Data Sources
- Local & Remote State

4. Terraform State Management

- Understanding Terraform State (terraform.tfstate)
- Local vs. Remote State
- Backends (S3, Azure Blob, GCS, Terraform Cloud)
- State Locking (DynamoDB for AWS)
- Importing Existing Resources
- State Manipulation (terraform state commands)





5. Terraform Modules

- What are Modules?
- Using Public Modules (Terraform Registry)
- Creating Custom Modules
- Module Composition & Best Practices
- Module Versioning

6. Advanced Terraform Features

- Dynamic Blocks
- Count & For-Each Meta-Arguments
- Conditional Expressions
- Workspaces (Managing Multiple Environments)
- Terraform Functions (join(), merge(), lookup(), etc.)
- Custom Validation Rules

7. Terraform with Cloud Providers

AWS

- Managing EC2, VPC, S3, IAM, RDS
- Using AWS SSM Parameter Store
- Managing EKS & ECS

Azure

- Managing VMs, VNet, Blob Storage, AKS
- Azure Key Vault Integration

Google Cloud (GCP)

- Managing Compute Engine, GKE, Cloud Storage
- IAM & Service Accounts

8. Terraform Security & Best Practices

- Securing Terraform Configs (Avoiding Secrets in State)
- Using tfvars & Environment Variables





- Remote Backend Security
- Policy as Code (Open Policy Agent - OPA, Sentinel)
- Terraform Compliance Testing

9. Terraform in CI/CD Pipelines

- Integrating Terraform with GitHub Actions
- GitLab CI/CD for Terraform
- Jenkins & Terraform
- Atlantis for Terraform Automation
- Terraform Cloud & Enterprise Features

10. Debugging & Troubleshooting

- Logging & Debugging (TF_LOG)
- Handling Dependency Errors
- Plan & Apply Failures
- Common Terraform Errors & Fixes

11. Terraform with Kubernetes

- Terraform Kubernetes Provider
- Managing Helm Releases with Terraform
- Deploying EKS, AKS, GKE

12. Terraform Advanced Patterns

- Multi-Cloud Deployments
- Blue-Green Deployments
- Zero-Downtime Deployments
- Managing Large-Scale Infrastructure

13. Terraform Cloud & Enterprise

- Terraform Cloud vs. Terraform Enterprise
- Remote Execution & Collaboration
- Cost Estimation





- Private Module Registry

14. Terraform with Other Tools

- Terragrunt (DRY Terraform)
- Packer + Terraform for Immutable Infrastructure
- Consul & Vault Integration

15. Real-World Projects & Case Studies

- Deploying a 3-Tier Architecture
- Multi-Region Deployment
- Serverless Deployment (Lambda, API Gateway)

16. Certification Preparation

- HashiCorp Certified: Terraform Associate Exam Topics
- Mock Tests & Practice Questions

Learning Path:

1. **Beginner:** Modules 1-5
2. **Intermediate:** Modules 6-10
3. **Advanced:** Modules 11-16

